

① Semi-latus rectum.

1. 양자

specific angular momentum.

$$\vec{h} = \vec{r} \times \vec{v}$$

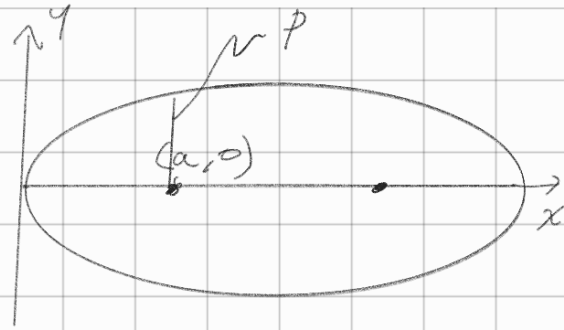
$$h = |\vec{h}|$$

$$p = \frac{h^2}{\mu}$$

semi-latus rectum

반동점

$(-a, 0)$



2. Unit

$$h = |\vec{r} \times \vec{v}|$$

$$[r] = \text{km}$$

$$[v] = \text{km/s}$$

$$\therefore [h] = \text{km}^2/\text{s}$$

$$[h^2] = \text{km}^4/\text{s}^2$$

$$[\mu] = \text{km}^3/\text{s}^2$$

$$\left[p = \frac{h^2}{\mu} \right] = \text{km.}$$

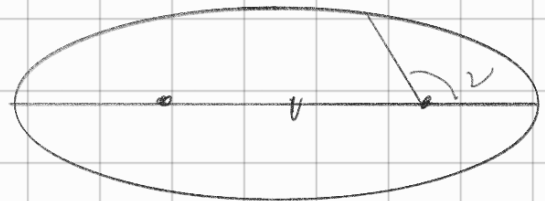
3. Meaning.

꺾는 양자 / orbit equation.

$$r = \frac{p}{1 + e \cos \nu}$$

$$p = \frac{h^2}{\mu}$$

($\nu = \text{true anomaly}$)



$\therefore p = \frac{h^2}{\mu}$, 꺾는 양자 , 꺾는 양자 관측 되는 궤도 파라미터

② Periapsis radius 근점반경

$$r = \frac{p}{1 + e \cos v}$$

if. $v = 0$

$$\cos v = \cos 0 = 1$$

$$r_p = \frac{p}{1 + e}$$

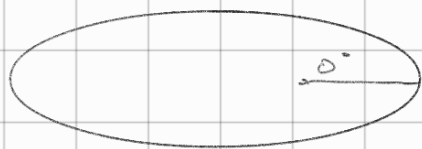
Use. $p = a(1 - e^2)$

at 근점

$$p = a(1 - e^2)$$

$$\begin{aligned} r_p &= \frac{a(1 - e^2)}{1 + e} \\ &= \frac{a(1 + e)(1 - e)}{(1 + e)} \end{aligned}$$

$$r_p = a(1 - e)$$



③ Apoapsis radius 원궤반경

$$r = \frac{p}{1 + e \cos v}$$

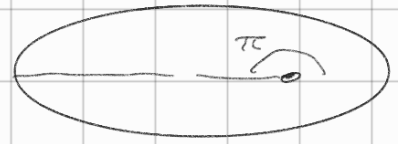
$$v = 180^\circ = \pi$$

$$\cos v = \cos \pi$$

$$r_a = \frac{p}{1 - e}$$

$$r_a = \frac{a(1 - e^2)}{1 - e}$$

$$r_a = a(1 + e)$$



④ Orbital period 궤도 주기

$$\frac{\mu}{r^2} = \frac{v^2}{r}$$

원궤도에서는 중력가속도가 구심가속도인 것이다.

$$\frac{\mu}{r} = v^2$$

$$v = \sqrt{\frac{\mu}{r}}$$

원궤도 둘레길이 $2\pi r$

$$T = \frac{2\pi r}{v}$$

$$= \frac{2\pi r}{\sqrt{\frac{\mu}{r}}}$$

$$= 2\pi r \sqrt{\frac{r}{\mu}}$$

$$= 2\pi \sqrt{\frac{r^3}{\mu}}$$

궤도에서 r 대신 a .

$$\therefore T = 2\pi \sqrt{\frac{a^3}{\mu}}$$

↳ Kepler's third law.

$$T^2 \propto a^3$$

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5)

Circular speed

원속도 속도.

원속도에서

중력가속도 = 원속도.

$$\frac{M}{r^2} = \frac{v_{\text{circ}}^2}{r}$$

$\times r$   $\rightarrow$   $\frac{M}{r} = v_{\text{circ}}^2$

$$\therefore v_{\text{circ}} = \sqrt{\frac{M}{r}}$$

⑥ 등속도 조건

비중속도 에너지

$$E = \frac{v^2}{2} - \frac{\mu}{r}$$

at parabolic escape boundary.

$$E = 0$$

$$0 = \frac{v_{esc}^2}{2} - \frac{\mu}{r}$$

$$v_{esc}^2 = \frac{2\mu}{r}$$

$$v_{esc} = \sqrt{\frac{2\mu}{r}}$$

